

Project Brief: Installation of an Urban Surveillance System

Project Overview

The installation of a public safety camera system represents a \$400,000 investment for Willow Creek, with deployment planned to begin on December 1, 2023. The project aims to significantly strengthen public safety through a strategically positioned camera network while respecting individual rights and supporting a broader smart, safe-town strategy.

Context and Rationale

Current Situation Assessment

Public safety statistics from recent years show a moderate but steady increase in nuisance behavior, vandalism, and minor offenses in certain areas of town. Resident surveys also indicate a growing feeling of insecurity, especially in the evening and in areas such as school surroundings, public squares, and parking lots. Local law enforcement resources are limited and cannot maintain a constant presence across the entire municipal area.

Legal and Ethical Framework

The project will comply with applicable federal, state, and local rules governing video surveillance in public spaces, including privacy protections, data security requirements, and any authorizations required by competent authorities. A dedicated ethics charter will define how the system may be used and will help safeguard fundamental rights.

Expected Benefits

- Reduce nuisance behavior and minor offenses by 15% to 20% in covered areas.
- Improve the sense of safety among residents and visitors.
- Support investigations by providing evidentiary material to authorized agencies.
- Optimize law enforcement response through better situational awareness.
- Improve protection of public facilities against vandalism.
- Contribute to traffic flow management and road safety.

Detailed Project Description

System Architecture

- 75 public safety cameras distributed across the municipal area.
- A Public Safety Monitoring Center equipped with video walls and operator workstations.
- A secure network infrastructure for data transmission.
- Redundant storage servers for temporary archiving of video footage.
- Next-generation video analytics software.
- Backup power systems to ensure continuity of service.

Equipment Types

- High-definition fixed cameras for sensitive locations.
- Pan-tilt-zoom dome cameras for broad-area coverage.
- Night-vision cameras for less illuminated areas.
- Automatic license plate recognition cameras at major town entrances.
- Temporary mobile cameras for special events.

Innovative Technologies

- Smart video analytics for automatic incident detection.
- End-to-end encryption of video streams to protect data.
- AI-assisted alert filtering to reduce false positives.
- Digital zoom capabilities with minimal quality loss.
- An intuitive management interface accessible from multiple authorized devices.

Strategic Deployment

Priority-Area Mapping

Camera placement was defined through a multi-criteria analysis that considered:

- Incident and offense statistics by area.
- High-traffic zones such as shops, schools, and transportation stops.
- Sensitive locations such as public buildings and parks.
- Major roads and intersections.
- Feedback from residents and business owners during consultations.

Coverage will be strengthened in:

- Downtown and shopping areas.
- School surroundings.
- Public parks and gardens.
- Municipal parking lots.
- Peripheral business areas.

Deployment Phasing

- Phase 1 (3 months): install the monitoring center and deploy 25 cameras downtown.
- Phase 2 (3 months): expand coverage around public facilities with 25 additional cameras.
- Phase 3 (2 months): cover peripheral areas and finalize deployment with 25 additional cameras.

Network Infrastructure

- Dedicated fiber network for high-density areas.
- Secure radio links for more remote areas.
- 4G/5G backup systems in case of outage.

Operational Organization

Public Safety Monitoring Center

- 1,290 sq ft (120 m²) facility in a secure municipal building.
- Latest-generation technical equipment.
- Video wall made up of 9 high-definition screens.
- 3 ergonomic operator workstations.
- Direct communication link with police and sheriff's office services.
- Strict access control supported by secure authentication.

Human Resources

- 6 trained video operators, with 2 per shift on rotation.
- 1 technical system manager.
- 1 administrative and legal supervisor.
- External technical support for specialized maintenance.

Operational Protocols

- Daily monitoring operations.
- Handling of detected incidents.
- Collaboration with law enforcement agencies.
- Extraction and transfer of footage under legally valid requests.
- Preventive and corrective maintenance of equipment.
- Management of outages and degraded-service situations.

Ethical and Legal Framework

Protection of Rights

- Automatic masking of private spaces that may fall within a camera’s field of view.
- A strict prohibition on filming inside homes.
- Clear signage informing the public where cameras are present.
- Access rights for individuals as provided by applicable law.
- Limited retention of recordings, with a maximum of 30 days unless legally required otherwise.
- Full logging and traceability of footage access.

Governance and Oversight

- Creation of an independent ethics committee including civil society representatives.
- Annual public report on system use.
- Regular technical and legal audits.
- Mandatory ethics training for operators.
- A dedicated professional conduct charter.

Financial Aspects

Detailed Budget

Item	Amount
Video equipment (cameras and mounts)	\$180,000
Secure network infrastructure	\$60,000
Monitoring center setup	\$50,000
Servers and storage systems	\$40,000
Software and licenses	\$35,000
Studies, design, and engineering	\$25,000
Staff training	\$10,000

Funding Plan

- Municipal budget: \$200,000 (50%).
- Federal or state public safety grant: \$120,000 (30%).
- State contribution: \$60,000 (15%).
- County contribution: \$20,000 (5%).

Operating Costs

Annual operating costs are estimated at \$120,000, including:

- Staffing costs: \$80,000.
- Technical maintenance: \$25,000.
- Energy consumption: \$5,000.
- Insurance and miscellaneous costs: \$10,000.

Integration into the Smart Town Strategy

Synergy with Other Municipal Projects

- Connection with traffic signal management to optimize traffic flow.
- Interface with smart street lighting to adjust lighting based on safety needs.
- Potential future smart city applications such as flow counting and environmental incident detection.
- Shared network infrastructure with other connected municipal services.

Technological Innovation

- Scalable platform that can integrate future technologies.
- Open architecture supporting interoperability with other municipal systems.
- Capacity to adapt to regulatory and technological changes.

Communication Plan

Pre-Deployment Communication

- Public information meetings by neighborhood.
- Special feature in the municipal magazine.
- Dedicated page on willowcreek.gov.
- Explanatory display at City Hall.
- Information brochures distributed to residents.

Ongoing Information

- Required public signage in covered areas.
- Regular updates on project progress.
- Communication of initial results and effectiveness statistics.
- Testimonials from local safety stakeholders and business owners.

Evaluation and Monitoring

Performance Indicators

- Statistical changes in offenses in covered areas.
- Number of incidents detected and handled through the system.
- Average response time after detection.
- Perception surveys among residents and business owners.
- Number of legally valid requests supported by footage.

Continuous Improvement Process

A steering committee will meet quarterly to analyze performance indicators, identify areas requiring adjustment, propose technical or organizational changes, and approve potential system extensions.

Conclusion

The installation of this public safety camera system is a significant step toward a safer and calmer Willow Creek. By combining advanced technology, strong respect for individual rights, and transparent governance, the project aims to balance public safety with the protection of fundamental rights. Beyond its safety dimension, it supports a broader smart town vision in which technology serves collective well-being and urban quality of life.